

TEST PLAN, QUALITY ASSURANCE & DEFECT ANALYSIS SPECIFICATION

For PharmaCart - Omni-Channel Pharmacy Ordering Platform

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Document Control & Metadata

- **Project Name:** PharmaCart — Online Ordering for Neighbourhood Pharmacy Chain
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- **Engineering Organization:** PharmaCart Project Delivery Team
- **Target Network:** 14 Retail Pharmacies across Ahmedabad, Gujarat, India
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Revision History

Version	Date	Author	Description of Change	Approved By
0.1	2026-1 0-02	Ritesh Jadhav	Initial test strategy and test case design	QA Team Lead
0.6	2026-1 0-04	Ritesh Jadhav	BVA, Decision tables, and equivalence partitioning	Software Architecture Lead
1.0	2026-1 0-05	Ritesh Jadhav	Execution defect log, DRE and Defect Density metrics	Head Pharmacist & Delivery Manager

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1. Test Strategy & Quality Objectives

The testing suite for PharmaCart enforces statutory patient safety, zero dispensing errors for Schedule H/X pharmaceuticals, and accurate synchronization with 14 disconnected retail POS installations across Ahmedabad.

Quality Objective	Operational Target Metric	Verification Method
Zero Schedule Drug Breaches	1 00% block of Schedule H/X drugs without verified Rx	Automated Regression & Negative Testing
Stock Accuracy at Counter	>= 98.0% fulfillment without stockout cancellation	Concurrency & Race-Condition Testing
Defect Removal Efficiency (DRE)	>= 90.0% defects filtered prior to production release	Defect Log Mathematical Analysis
Defect Density	<= 0.30 Defects / Story Point (<= 2.0 Defects / KLOC)	Codebase & Defect Tracking Audit
Verification SLA Adherence	90% of prescriptions audited within <= 1 5 minutes	Performance & Latency Stress Testing

2. Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) targets the edges of input domains where software bugs cluster due to relational operator errors (< vs <= , or > vs >=) .

2.1 Order Quantity Input Boundary Testing (Min: 1, Max: 30)

- Domain Specification:** A customer may purchase a minimum of **1 unit** (single strip/bottle) up to a statutory retail maximum of **30 units** per order to prevent hoarding and illicit bulk redistribution.
- Input Domain:** Valid Integer range [1, 30].
- Testing Values:**
 - Minimum Boundary: Min - 1 = 0, Min = 1, Min + 1 = 2.
 - Nominal Value: Nominal = 15.
 - Maximum Boundary: Max - 1 = 29, Max = 30, Max + 1 = 31.
 - Extreme Robustness Values: -1, 100, Non-numeric ("ten") , Decimal (2.5) .

Test Case ID	Test Parameter	Input Value	Boundary Category	Expected System Behavior	Actual Result	Status
TC-B VA-QT Y-01	Order Quantity	-1	Extreme Lower Invalid	Block input; display: "Quantity must be at least 1"	Blocked with validation error	PASS
TC-B VA-QT Y-02	Order Quantity	0	Lower Boundary (Min - 1)	Block input; cart rejects zero items	Blocked; cart button disabled	PASS
TC-B VA-QT Y-03	Order Quantity	1	Lower Boundary (Min)	Accept; minimum permissible order line item	Added to cart (Qty = 1)	PASS
TC-B VA-QTY -04	Order Quantity	2	Lower Boundary (Min + 1)	Accept; standard multi-unit purchase	Added to cart (Qty = 2)	PASS
TC-B VA-QT Y-05	Order Quantity	15	Nominal Valid Value	Accept; typical monthly chronic prescription	Added to cart (Qty = 15)	PASS
TC-B VA-QT Y-06	Order Quantity	29	Upper Boundary (Max - 1)	Accept; high-volume retail purchase	Added to cart (Qty = 29)	PASS
TC-B VA-QT Y-07	Order Quantity	30	Upper Boundary (Max)	Accept; maximum permissible retail order	Added to cart (Qty = 30)	PASS
TC-B VA-QTY -08	Order Quantity	31	Upper Boundary (Max + 1)	Block input; display: "Maximum allowed retail limit is 30 units"	Blocked; alert rendered	PASS
TC-B VA-QT Y-09	Order Quantity	2.5	Fractional / Decimal	Block input; integer enforcement rejects fractional strips	Auto-rounded or blocked	PASS
TC-B VA-QT Y-10	Order Quantity	"ABC"	Non-numeric String	Sanitization rejects string; retains previous state	Rejected; field reset to 1	PASS

2.2 Coupon Value & Discount Threshold Boundary Testing

- **Domain Specification:**
 - Cart minimum subtotal threshold for coupon eligibility: **₹500.00**.
 - Discount calculation: **10% of OTC items**, subject to a maximum cap of **₹100.00**.
 - Absolute statutory rule: Prescription (Schedule H/X) items are excluded from discount computations.
- **Testing Values for Cart Subtotal:**
 - Cart Min Boundary: ₹499.00 (Min - 1) , ₹500.00 (Min), ₹501.00 (Min + 1) .
- **Testing Values for Calculated Discount:**
 - Discount Cap Boundary: ₹990 cart (10% = ₹99.00), ₹1,000 cart (10% = ₹100.00), ₹1,010 cart (10% = ₹101.00 capped to ₹100.00).

Test Case ID	Cart Subtotal (OTC)	Coupon Code	Boundary Category	Expected System Behavior	Actual Result	Status
TC-BVA-CPN-01	₹499.00	PHARMA10	Subtotal Boundary (Min - 1)	Reject coupon; "Add ₹1 .00 more to apply coupon"	Coupon rejected; message displayed	PASS
TC-BVA-CPN-02	₹500.00	PHARMA10	Subtotal Boundary (Min)	Accept coupon; discount = ₹50.00 (1 0% of ₹500)	Applied discount: ₹50.00	PASS
TC-BVA-CPN-03	₹501 .00	PHARMA10	Subtotal Boundary (Min + 1)	Accept coupon; discount = ₹50.1 0	Applied discount: ₹50.1 0	PASS
TC-BVA-CPN-04	₹990.00	PHARMA10	Discount Cap (Cap - 1)	Accept coupon; discount = ₹99.00	Applied discount: ₹99.00	PASS
TC-BVA-CPN-05	₹1 , 000.00	PHARMA10	Discount Cap (Cap)	Accept coupon; discount = ₹1 00.00 (Exact max cap)	Applied discount: ₹1 00.00	PASS
TC-BVA-CPN-06	₹1 , 500.00	PHARMA10	Discount Cap (Cap + 1)	Accept coupon; discount capped at exactly ₹1 00.00	Applied discount: ₹1 00.00	PASS
TC-BVA-CPN-07	₹800 (Rx only)	PHARMA10	Schedule Drug Exclusion	Reject coupon; "Coupons apply exclusively to OTC products"	Coupon rejected	PASS

3. Equivalence Class Partitioning (ECP)

Equivalence Partitioning divides input data into classes of equivalent data from which test cases can be derived, assuming all members of a partition are processed identically.

Partition ID	Input Field / Parameter	Valid Equivalence Classes (VEC)	Invalid Equivalence Classes (IEC)
ECP-01	Order Quantity	VEC-1 : Integers in [1 , 30]	IEC-1 : Integers < 1 IEC-2: Integers > 30 IEC-3: Floats / Decimals IEC-4: Non-numeric characters
ECP-02	Prescription File Upload	VEC-2: Formats: JPEG, PNG, PDF VEC-3: File Size <= 1 0 MB	IEC-5: Executables (.exe, .sh, .bat) IEC-6: Corrupted images (0 bytes) IEC-7: File Size > 1 0 MB
ECP-03	Prescription Issue Date	VEC-4: Date issued within <= 30 days	IEC-8: Date issued > 30 days (Expired) IEC-9: Future date (> Today)
ECP-04	Store Physical Stock	VEC-5: Physical Count > 2 units (Available > 0)	IEC-1 0: Physical Count <= 2 units (Buffer Lock) IEC-1 1 : Physical Count = 0 (Total Stockout)
ECP-05	Customer Pickup OTP	VEC-6: Exactly 6 numeric digits matching hash	IEC-1 2: String length != 6 IEC-1 3: Alphanumeric / Special characters IEC-1 4: Expired OTP (> 24 hours)

4. Comprehensive Decision Table: Prescription & Stock Rules

4.1 Condition & Action Definitions

To test the complex combinatorial business logic governing prescription compliance and multi-store inventory reservation, we construct a formal Decision Table.

Conditions (Inputs):

- **C1: Schedule Drug?** Does the basket contain at least one Schedule H, H1, or X medication? (Y / N)
- **C2: Prescription Uploaded?** Has the customer provided a readable doctor prescription document? (Y / N)
- **C3: Pharmacist Approved?** Has a licensed pharmacist validated doctor seal, reg number, and approved the order? (Y / N)
- **C4: Store Stock Available?** Does the target store have sufficient stock after applying the 2-unit safety buffer ($Physical - 2 \geq OrderQty$)? (Y / N)

Actions (System Outputs):

- **A1: Allow Instant Checkout.** Proceed directly to checkout without pharmacist gate.
- **A2: Prompt Prescription Upload.** Block checkout and render mandatory upload screen.
- **A3: Enqueue to Pharmacist Desk.** Route ticket to central FIFO tele-pharmacy verification portal.
- **A4: Reject Order with Clinical Alert.** Cancel order, notify customer of rejection reason, and provide re-upload link.
- **A5: Soft Reserve Stock (30 Mins).** Place 30-minute reservation lock on store POS inventory.
- **A6: Route to Sister Branch.** Automatically re-route order to the next closest Ahmedabad store with available stock.
- **A7: Issue Picking Slip & OTP.** Confirm order, alert store staff to pack, and generate customer pickup OTP.

4.2 Tabular Decision Truth Matrix

The $2^4 = 16$ possible logical permutations are collapsed into 8 discrete, mutually exclusive business decision rules.

Rule Category	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Condition Evaluation								
C1 : Schedule H/X Drug in Cart?	N	N	Y	Y	Y	Y	Y	Y
C2: Prescription Uploaded?	—	—	N	Y	Y	Y	Y	Y
C3: Pharmacist Approved?	—	—	—	N	Y	Y	Y	Y
C4: Stock Available (Phys - 2 >= Qty)?	Y	N	—	—	Y	N (Local)	N (All 1 4)	Y (High Vol)
Action Execution								
A1 : Allow Instant Checkout	X							
A2: Prompt Prescription Upload			X					
A3: Enqueue to Pharmacist Desk					X	X	X	X
A4: Reject Order with Clinical Alert				X				
A5: Soft Reserve Stock (30 Mins)	X				X			X
A6: Route to Sister Branch		X				X		
A7: Issue Picking Slip & OTP	X				X			

Expected System Behavior	OTC Fast Checkout	OTC Sister	Rx Blocked at Cart	Rx Legally Rejected	Standard Rx Approved	Sister Store Failover	Chain-wide	Split-Order Exception
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Rule Category	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
		Re-route					Out of Stock	

4.3 Test Case Derivation from Decision Rules

- **Test Case realization for Rule 1 (OTC Fast Path):** Customer adds 2 strips of Paracetamol (OTC) at Navrangpura store; stock is 10 ($10 - 2 = 8 \geq 2$). Cart proceeds directly to payment without prompting for prescription.
- **Test Case realization for Rule 4 (Prescription Rejected):** Customer orders Amoxicillin 500mg (Schedule H) and uploads prescription dated 45 days ago. Pharmacist clicks "Reject: Expired Rx". Order immediately transitions to `RX_REJECTED` ; zero store inventory is locked.
- **Test Case realization for Rule 6 (Sister Store Re-route):** Customer orders Metformin 500mg at Satellite branch; store physical stock is 3 ($3 - 2 = 1 < 2 \text{ ordered}$). System detects shortage, verifies Bodakdev store (2.1 km away) has 20 units, and automatically reallocates order to Bodakdev.

5. Systematic Test Case Suite

The test cases below validate end-to-end trace realization linking directly to the SRS requirements matrix.

Test Case ID	Requirement Ref	Test Description & Objective	Test Input Data	Expected Result	Execution Result
TC-RX-01	FR-01	Upload valid prescription image (PNG)	2.4 MB clear JPG prescription	Prescription encrypted AES-256; unique Rx_ID generated	PASS
TC-RX-02	FR-01	Upload malicious executable payload	exploit.pdf.exe (Renamed PE)	File type scanner detects MIME mismatch; blocks upload	PASS
TC-VR-01	FR-02	Pharmacist approval workflow with Council ID	Valid Rx; Pharmacist PIN = 4491	Approval stamped with Council ID GPC-44910 ; order unblocked	PASS
TC-VR-02	FR-02	Pharmacist rejection with audit reason code	Blurred Rx image	Rejection recorded; SMS alert dispatched to customer	PASS
TC-INV-01	FR-04	Ingest 1 4-store delta POS sync stream	JSON stream from 1 4 store clients	Central stock ledger updated in <= 30 seconds	PASS
TC-BUF-01	FR-05	Enforce 2-unit safety buffer rule	Physical Stock = 2; Order Qty = 1	System reports "Out of Stock" (Effective stock = 0)	PASS
TC-BUF-02	FR-05	30-minute soft reservation timeout	Unpaid order held for 30m 01 s	Soft lock released automatically; shelf units restored	PASS
TC-RT-01	FR-06	Multi-store geodesic distance routing	Delivery to Pincode 380009	Routed to Navrangpura store (Distance = 1 .2 km)	PASS
TC-OTP-01	FR-08	Counter pickup verification with valid OTP	Correct 6-digit OTP entered	Terminal confirms match; prints GST tax invoice	PASS
TC-OTP-02	FR-08	Brute-force protection on pickup OTP	3 consecutive incorrect OTPs	Terminal locks OTP entry for 1 5 minutes; alerts store manager	PASS

6. Comprehensive Defect Log (Execution Results)

The defect log records all software defects identified, classified, and resolved during Sprint 4 and Sprint 5 testing cycles across the 14-store testbed.

Defect ID	Severity	Module / Component	Defect Description	Root Cause	Status	Reso Sp
BUG-01	Critical	ReservationManager	Race condition allowed 2 online orders for last 3 units concurrently	Soft hold lacked database row-level locking (SELECT FOR UPDATE)	Closed	Sp1
BUG-02	Critical	PrescriptionService	Prescriptions uploaded without doctor registration number were accepted	Regex parser failed on non-standard state council registration prefixes	Closed	Sp1
BUG-03	Major	StoreSyncDaemon	POS sync daemon crashed when store Windows terminal rebooted	Daemon lacked auto-restart Windows Service wrapper	Closed	Sp1
BUG-04	Major	PromotionEngine	1 0% coupon applied across both OTC and Schedule H prescription items	Subtotal query failed to filter by is_prescription_mandatory = false	Closed	Sp1
BUG-05	Major	RoutingEngine	Sister store routing failed if customer address lacked latitude/longitude	Geocoder fallback to Ahmedabad 6-digit pincode was unhandled	Closed	Sp1
BUG-06	Medium	PharmacistPortal	Dual-pane image viewer failed to rotate vertical mobile photos	EXIF orientation metadata was stripped during WebP compression	Closed	Sp1
BUG-07	Medium	AuthService	Pickup OTP SMS delayed by > 5 minutes during peak evening hours	Karix SMS gateway priority queue was set to promotional instead of transactional	Closed	Sp1
BUG-08	Medium	OrderService	Cancelled order failed to release inventory lock on Store #07 POS	Missing exception handler on payment gateway cancellation webhook	Closed	Sp1
BUG-09	Minor	CustomerUI	Cart badge did not increment dynamically without page refresh	React query cache invalidator missing on addToCart event	Closed	Sp1

BUG-10	Minor	StoreTerminal	Thermal receipt printer cut line	Receipt template width configured for 58mm	Closed	Sp1
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Defect ID	Severity	Module / Component	Defect Description	Root Cause	Status	Reso Sp
			truncated patient doctor name	instead of 80mm roll		
BUG-11	Major	ReservationManager	Safety buffer deducted twice when order was edited in cart	Idempotency token was regenerated on cart quantity update	Closed	Sp1
BUG-12	Medium	CatalogService	Search query for generic "Paracetamol" failed to return "Crocina"	Full-text search index lacked pharmaceutical synonym dictionary	Closed	Sp1
BUG-13	Minor	CustomerUI	Date-picker allowed selecting past birthdates greater than 120 years	Missing boundary validation on patient profile age field	Closed	Sp1
BUG-14	Critical	PaymentService	User double-clicking "Pay" charged credit card twice	UI submit button lacked debounce handler; backend lacked idempotency key	Closed	Sp1
BUG-15	Medium	LogisticsService	Runner manifest listed delivery addresses in random non-geodesic sequence	Missing traveling-salesperson route optimization sort	Closed	Sp1
BUG-16	Major	PrescriptionService	PDF prescriptions containing multiple pages only uploaded page 1	PDF-to-image converter failed to loop through document page array	Closed	Sp1
BUG-17	Minor	StoreTerminal	Barcode scanner beeped error on valid EAN-13 barcodes with leading zeros	String parser trimmed leading zero character	Closed	Sp1
BUG-18	Medium	NotificationService	WhatsApp notification template failed for non-English customer names	Character encoding set to ASCII instead of UTF-8	Closed	Sp1

7. Quantitative Defect Metrics & Formulas

7.1 Defect Density Computation (per Story Point & KLOC)

Defect Density measures the residual defects relative to software size, providing an objective metric of architectural quality and test thoroughness.

Metric Definitions:

- **Total Defects Discovered in Testing (D_{internal}):** 18 Defects.
- **Core MVP Scope Size:** 72 Story Points.
- **Codebase Size (KLOC):** 12.5 Thousand Lines of Code (KLOC) across microservices.

Calculation 1: Defect Density per Story Point

- **Formula:**

$$\text{Defect Density (per Story Point)} = \text{Total Internal Defects} / \text{Total Story Points}$$

- **Step-by-Step Calculation:**

$$\text{Defect Density (SP)} = 18 / 72 = 0.25 \text{ Defects / Story Point}$$

Calculation 2: Defect Density per KLOC

- **Formula:**

$$\text{Defect Density (per KLOC)} = \text{Total Internal Defects} / \text{Codebase Size in KLOC}$$

- **Step-by-Step Calculation:**

$$\text{Defect Density (KLOC)} = 18 / 12.5 = 1.44 \text{ Defects / KLOC}$$

7.2 Defect Removal Efficiency (DRE) Computation

Defect Removal Efficiency (DRE) quantifies the quality team's ability to filter out defects prior to customer exposure.

Metric Definitions:

- **E (Internal Defects):** Defects caught and resolved by the QA engineering team prior to production deployment = **18 Defects**.
- **D (Escaped Defects):** Defects reported by real users / store clerks during the initial 3-store production pilot in Ahmedabad = **2 Defects** (Defect 1 : Minor UI alignment on low-res POS monitor; Defect 2: Thermal printer font size).

DRE Formula:

$$\text{DRE (\%)} = [E / (E + D)] * 100$$

Step-by-Step Intermediate Calculation:

```
Step 1: Calculate Total Identified Defects
Total_Defects = E + D = 18 + 2 = 20 Defects

Step 2: Compute Removal Quotient
DRE_Quotient = 18 / 20 = 0.9000

Step 3: Convert to Percentage
DRE = 0.9000 * 100 = 90.0%
```

7.3 Benchmark Evaluation & Quality Gate Assessment

Quality Metric	Measured Value	Industry Standard Benchmark	Compliance Assessment
Defect Removal Efficiency (DRE)	90.0%	85.0% – 90.0%	EXCELLENT: Meets high-reliability industry benchmark.
Defect Density (per KLOC)	1.44	1.00 – 3.00 Defects/KLOC	OPTIMAL: Indicates clean code architecture and low technical debt.
Defect Density (per Story Point)	0.25	0.20 – 0.40 Defects/SP	CONTROLLED: Sits well within acceptable commercial thresholds.
Critical Defect Escapes	0	Strict 0 Tolerance	PERFECT: Zero regulatory or medical safety defects escaped to pilot.

Formal Quality Assurance Sign-Off

The undersigned confirm that the testing procedures, boundary analyses, decision models, and mathematical defect evaluations documented herein have been executed in compliance with IEEE Std 829. Release 1.0 achieves all required quality thresholds for chain-wide commercial launch.

Executed & Documented By:

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